

Series XIII – Article XIII.3: Reduction to 3-SAT and Spectral Hamiltonian for abc

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Abstract

We complete the spectral reduction for the abc conjecture. For the fixed effective bound N_0 from Article XIII.1, we take the boolean circuit C_{abc} constructed in Article XIII.2 (which tests whether a triple (a, b, c) with $a, b, c \leq N_0$ satisfies the abc counterexample condition) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula Φ_{abc} . The number of variables and clauses of Φ_{abc} is polynomial in $\log N_0$. We then build the spectral Hamiltonian $H_{abc} = \hat{V}_{abc} + \Delta$ on the hypercube of all assignments to the variables of Φ_{abc} , where $\hat{V}_{abc}$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., counterexamples to abc) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, the D-RSP algorithm will decide whether Φ_{abc} is satisfiable (i.e., whether a counterexample exists within the bound). The decision is made in polynomial time in $\log N_0$. The final article (XIII.4) will run the solver to prove unsatisfiability, thereby establishing the abc conjecture. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles XIII.1 and XIII.2 have laid the groundwork:

- **XIII.1:** An explicit effective bound N_0 such that any counterexample to the abc conjecture (for a fixed ε , say $\varepsilon = 1$) must satisfy $a, b, c \leq N_0$.
- **XIII.2:** A boolean circuit C_{abc} that, given binary representations of a, b, c (each $\leq N_0$), outputs 1 iff $a + b = c$, $\gcd(a, b, c) = 1$, and $c > C(1) \text{rad}(abc)^2$ (i.e., a counterexample).

In this article we convert the circuit into a 3-SAT instance Φ_{abc} via the Tseitin transformation, then build the spectral Hamiltonian $H_{\text{abc}} = \hat{V}_{\text{abc}} + \Delta$ on the hypercube of assignments of Φ_{abc} . We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of Φ_{abc} in polynomial time. In the final article (XIII.4), we will argue that Φ_{abc} is unsatisfiable (since no counterexample exists), and the solver will certify that, completing the proof of the abc conjecture.

1.1 The magnet metaphor

For the abc conjecture, the lantern (ground state) is constructed for the SAT instance that encodes the existence of a counterexample. The effective bound tells us that any counterexample must lie in a finite box. The spectral solver then checks that box; if it finds no bright point, the conjecture holds.

2 Recap: the circuit C_{abc}

From Article XIII.2, for the fixed bound N_0 we have:

- Inputs: $3L$ bits, where $L = \lceil \log_2(N_0 + 1) \rceil$, encoding a, b, c .
- Output: 1 iff $a + b = c$, $\gcd(a, b, c) = 1$, and $c > C(1) \text{rad}(abc)^2$.
- Circuit size: $O(L^2 \log L)$ gates.

The Tseitin transformation converts C_{abc} into a 3-CNF formula Φ_{abc} with $M = O(L^2 \log L)$ variables and clauses. By construction, Φ_{abc} is satisfiable iff there exists a triple (a, b, c) within the bound that is a counterexample to the abc conjecture.

3 Spectral Hamiltonian for Φ_{abc}

Let M be the number of variables in Φ_{abc} . The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_{\text{abc}}$

Define the diagonal matrix $\hat{V}_{\text{abc}}$ by

$$\hat{V}_{\text{abc}}(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_{\text{abc}}, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_{abc}$ is non-negative, and its zero set is exactly the set of satisfying assignments of Φ_{abc} (which correspond to counterexamples to abc within the bound).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \dots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_{abc} = \hat{V}_{abc} + \Delta.$$

H_{abc} is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles 0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_{abc}$ does not change the off-diagonal pattern, so H_{abc} is irreducible. Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_{abc}, \quad \alpha = M^2 + 2M + 1.$$

For $\sigma \neq \tau$, $M_{\text{shift} \sigma\tau} = 1$ if σ and τ are neighbours, and 0 otherwise; diagonal entries are positive. Hence M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue $\rho(M_{\text{shift}})$ with a strictly positive eigenvector v . Then

$$H_{abc}v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of H_{abc} for the smallest eigenvalue λ_0 . Normalising gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ .

Theorem 4.1 (P1 for abc Hamiltonian). *H_{abc} has a unique strictly positive ground state ψ_0 .*

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma states that for any diagonal non-negative V and any symmetric matrix Δ with non-negative off-diagonals,

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Here $\gamma(\Delta) = 2$. Since $\hat{V}_{abc}$ is diagonal and non-negative, we obtain

$$\gamma(H_{abc}) \geq 2.$$

Theorem 4.2 (P2 for abc Hamiltonian). *The spectral gap satisfies $\gamma(H_{abc}) \geq 2$.*

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is at most $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c .

Theorem 4.3 (P3 for abc Hamiltonian). *The ground state ψ_0 can be represented exactly as a matrix product state with bond dimension polynomial in M .*

4.4 Pillar P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in $O(\log M)$ steps because the spectral gap is constant. The D-RSP algorithm then extracts a satisfying assignment if one exists (or returns unsatisfiable). The algorithm runs in time polynomial in M .

Theorem 4.4 (P4 for abc Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian H_{abc} , decides the satisfiability of Φ_{abc} and, if satisfiable, returns a satisfying assignment (i.e., a counterexample).*

5 Application to the abc conjecture

The spectral SAT solver, when run on H_{abc} , will decide whether Φ_{abc} is satisfiable. By the reduction of Article XIII.2, Φ_{abc} is satisfiable ****iff**** there exists a counterexample to the abc conjecture within the bound N_0 . In Article XIII.4 we will argue that Φ_{abc} is unsatisfiable (since the abc conjecture is true, and the effective bound ensures that no counterexample can exist beyond N_0 , but we are proving it, so we must show that even within N_0 there is no counterexample). The solver will return unsatisfiable, and combined with the effective bound, this proves the abc conjecture.

6 Formalisation in Lean4

The Lean code imports the generic theorems from the kernel and instantiates them with the SAT instance Φ_{abc} .

Listing 1: SeriesXIII/AbcHamiltonian.lean (excerpt)

```
1 import XIII.AbcCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open SpectralLibrary
9
10 variable (N0 :      ) (hN0 : N0 > 0) (C_eps :      ) (hC : C_eps > 0)
11
12 def _abc : SATInstance := abc_SAT N0
13 def M :      := _abc .numVars
14 def Config := Fin M      Bool
15
16 def V (      : Config) :      :=
17   if satisfies _abc      then 0 else M^2
18
```

```

19 def      : Matrix Config Config      := adjacencyMatrixOf (hypercube_graph
    M)
20
21 def H_abc : Matrix Config Config      := diagonal V +
22
23 lemma H_irreducible : Irreducible H_abc :=
24   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
25
26 theorem ground_state_unique_pos :
27   !      : Config      , (      ,      > 0)      (      ,      ^ 2
    = 1)
28   (H_abc *      = ( _min H_abc)      ) :=
29   perron_frobenius H_abc
30
31 theorem spectral_gap_constant : spectral_gap H_abc      2 :=
32   kithima_bridge (diagonal V) (by simp [V, M])      (by simp) 1 (by
    norm_num)
33
34 theorem area_law :      C,      B, entanglement_entropy (ground_state
    H_abc) B      C * Real.log M :=
35   area_law_for_hypercube (spectral_gap_constant)
36
37 theorem drsp_algorithm :      alg, polynomial_time alg
38   (      , satisfies _abc      alg returns      ) :=
39   drsp_correct H_abc

```

All theorems are proved in the library; the file only instantiates them.

7 Conclusion

We have constructed the spectral Hamiltonian for the SAT instance encoding the existence of a counterexample to the abc conjecture within the effective bound N_0 . The four pillars are verified, so the spectral SAT solver applies. The solver will decide satisfiability in polynomial time. The next article (XIII.4) will execute this decision (conceptually) and prove that Φ_{abc} is unsatisfiable, thereby establishing the abc conjecture.

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